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Intelligent Traffic Path Recommendations Using Machine Learning based Temporal Feature Aggregation and Time Series Based Graph Routing

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ABSTRACT

Urban transportation networks generate massive volumes of spatio-temporal data, and the effective exploitation of such data to enable intelligent route recommendations has become a critical priority in smart city development. Existing systems predominantly rely on static or short-horizon prediction models that fail to capture the complex, non-stationary dynamics characterizing real-world vehicular traffic, particularly under congestion-prone conditions in densely populated urban corridors. The challenges are further compounded by high-dimensional, noisy, and heterogeneous sensor data that makes robust feature extraction and temporal modeling particularly difficult, as traditional approaches exhibit poor generalization across varying prediction horizons and traffic regimes. This research proposes an integrated framework for traffic path recommendation that combines systematic feature extraction, temporal prediction, and dynamic graph-based routing. The methodology begins with the Feature Extraction for Vehicular Traffic Time Series (FEVTTS) algorithm, which applies Mutual Information, Random Forest Regression, and Correlation Coefficient analysis to derive an optimal, low-redundancy feature set from raw sensor data. The refined features are then fed to the Temporal Traffic Feature Aggregation Algorithm (TFTAA), which integrates short-term and long-term traffic patterns using a rolling window mechanism to produce accurate multi-horizon forecasts. Finally, the predicted traffic variables are embedded as dynamic edge weights into a Time Series Based Graph (TSBG), on which Dijkstra's algorithm computes optimal time-dependent routes. The proposed TFTAA algorithm employs a hybrid statistical and ensemble machine learning design, incorporating SARIMAX for seasonal decomposition and gradient-boosted residual correction using XGBoost, augmented by quantile regression for uncertainty quantification. The framework was validated using the London Traffic Dataset (Highways England) and a locally collected Pune city dataset comprising 26,785 and 25,920 samples, respectively, across diverse road types. TFTAA consistently outperformed SARIMAX and XGBoost baselines at all evaluated prediction horizons from 5 to 60 minutes, achieving a Mean Absolute Error (MAE) of 2.1 at the 5-minute horizon and 14.3 at the 60-minute horizon, compared to 25.1 for SARIMAX at the same horizon. The TSBG-based routing reduced average travel time by an average of 18.3% over static Dijkstra across 15 test days on the London dataset. The proposed framework demonstrates that tightly coupling predictive traffic modeling with dynamic graph routing yields measurable improvements in route efficiency over baseline approaches. The research advances the state of the art in intelligent transportation systems by

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Keywords - traffic path recommendation, time series forecasting, TFTAA algorithm, feature extraction, TSBG routing, machine learning

providing a scalable, feature-engineered, and prediction-integrated path recommendation solution applicable to both developed and emerging urban road networks.

1. Introduction

The rapid proliferation of smart transportation technologies has transformed urban road networks into data-rich environments where millions of vehicle interactions are captured every hour. Among the most consequential applications of this data is traffic path recommendation — the problem of determining optimal routes for vehicles navigating complex, congested urban networks. Unlike classical shortest-path algorithms that operate on static road graphs [1], contemporary systems must account for the dynamic, time-dependent nature of traffic conditions, which are governed by daily and seasonal demand patterns, weather, incidents, and unpredictable behavioral factors [2]. India's road transport network, the second largest in the world by area, exemplifies the challenges of large-scale traffic management. National Highways carry the greatest proportion of vehicular load, yet persistent congestion due to rapid urbanization and inadequate infrastructure leads to substantial economic and environmental costs [3]. Path recommendation algorithms that merely minimize physical distance fail to account for prevailing congestion dynamics, resulting in suboptimal routing that increases travel time and emissions. The growing deployment of GPS-enabled devices, IoT roadside sensors, inductive loop detectors, and mobile crowd-sourcing platforms such as Waze and Google Maps has made high-frequency, multi-modal traffic time series data available at unprecedented scale [4].

Time series analysis of traffic data reveals strong temporal structure: peak-hour patterns, weekday-versus-weekend variations, and seasonal demand cycles are consistently observable in flow, speed, and occupancy measurements [5]. Statistical models such as ARIMA and SARIMAX have traditionally been employed to exploit this structure, offering interpretable and computationally efficient forecasts for short-to-medium horizons [6]. However, these models assume stationarity and linearity, assumptions that are frequently violated in congested urban traffic where non-linear phase transitions between free-flow and stop-and-go conditions occur rapidly and unpredictably [7]. Machine learning methods, particularly ensemble approaches such as XGBoost and gradient-boosted decision trees, have demonstrated strong performance in non-linear traffic prediction tasks when paired with well-engineered feature inputs [8]. The key insight driving their superiority over classical statistical models is their ability to exploit complex feature interactions without imposing linearity constraints. Feature selection and extraction thus become critical upstream processes: supplying redundant, correlated, or uninformative features to an ensemble model significantly degrades its predictive accuracy and generalization capacity [9].

Beyond predictive modeling, the effective integration of traffic forecasts into routing algorithms is a distinct and equally important research challenge. Graph-based routing algorithms such as Dijkstra's, A*, and AO* traditionally operate on static cost graphs where edge weights represent free-flow travel times or distances [10]. Adapting these algorithms to dynamic, prediction-informed graphs requires a principled framework for mapping time-indexed traffic forecasts onto edge weights in a way that enables time-dependent shortest-path queries. The Time Series Based Graph (TSBG) formulation proposed in this research directly addresses this requirement by defining edge weights as functions of predicted traffic variables at the time of traversal. This paper addresses the complete pipeline from raw sensor data to route recommendation through three algorithmic contributions: (i) the Feature Extraction for Vehicular Traffic Time Series (FEVTTS) algorithm, which derives an optimal low-dimensional feature set using Mutual Information, Random Forest, and Correlation Coefficient methods; (ii) the Temporal Traffic Feature Aggregation Algorithm (TFTAA), which employs a hybrid SARIMAX-

gradient boosting architecture with rolling window updates to produce accurate multi-horizon traffic forecasts; and (iii) the TSBG-based routing module, which embeds TFTAA predictions as dynamic edge weights for time-dependent Dijkstra routing. Empirical validation on the London Traffic Dataset and a locally collected Pune urban dataset demonstrates the superiority of the proposed approach over SARIMAX and XGBoost baselines across all evaluated prediction horizons and routing efficiency metrics.

2. Literature Review

2.1 Traffic Time Series Forecasting: Classical Methods

The foundations of traffic time series forecasting were established by Box-Jenkins autoregressive integrated moving average (ARIMA) models, introduced to transportation applications in the late 1970s. Ahmed and Cook [41] demonstrated that ARIMA-based approaches could capture the temporal autocorrelation inherent in freeway flow data, yielding short-horizon predictions competitive with naive models. The key advantage of ARIMA lies in its interpretability and modest data requirements relative to machine learning approaches, making it attractive for operational deployment in data-scarce settings. However, its stationarity assumption and inability to model non-linear dynamics imposed fundamental accuracy ceilings in congested corridors. Seasonal ARIMA (SARIMA) extended the Box-Jenkins framework by incorporating multiplicative seasonal operators that model nested temporal periodicities (daily, weekly, annual) characteristic of transportation demand. Dubey et al. [95] and Sirisha et al. [96] demonstrated empirically that SARIMA consistently outperforms non-seasonal ARIMA on strongly periodic traffic series, effectively reducing forecasting bias across mid-range horizons. The SARIMAX variant further augments the model with exogenous regressors, enabling the incorporation of external covariates such as weather indicators and calendar effects that modulate traffic demand. This model forms a baseline component of the hybrid framework proposed in this study. Davis and Nihan [142] introduced non-parametric regression for short-term freeway forecasting, demonstrating that kernel-smoothing approaches can accommodate non-stationary traffic without explicit model specification. Zahid et al. [143] applied quantile forest regression for probabilistic short-term travel speed forecasting, extending point estimates to distributional predictions that support risk-aware routing. These probabilistic extensions are particularly relevant for the proposed framework, which requires uncertainty-aware edge weight assignment in the TSBG routing module.

2.2 Machine Learning Approaches

The transition from classical statistical models to machine learning brought significant improvements in capturing the non-linear dynamics of urban traffic. Lin et al. [53] demonstrated that Support Vector Regression (SVR) combined with Maximal Information Coefficient (MIC) feature selection achieved competitive short-term prediction accuracy with substantially lower computational overhead than deep learning models, making SVR a practical choice for resource-constrained deployments. Sun et al. [56] applied XGBoost with spatio-temporal segmentation on automatic number plate recognition data, showing that gradient-boosted ensembles naturally handle missing sensor values through split-based imputation while producing interpretable feature importance rankings. Abduljabbar et al. [57] evaluated traffic prediction on arterial roads using both traffic and weather features, confirming that multi-modal feature engineering consistently reduces prediction error across ensemble methods. Alsolami et al. [58] provided a structured review of hybrid statistical and machine learning methods, concluding that ensemble models achieve their best performance when combined with principled feature selection rather than operating on raw or lightly processed inputs. This finding directly motivates the FEVTTS feature extraction layer proposed in the current framework. Pirani et al. [59] conducted a comparative study of ARIMA, GRU, LSTM, and BiLSTM on time-series prediction, demonstrating that gradient boosting methods can outperform recurrent deep learning models when features are carefully crafted, particularly under data-limited conditions. The Random Forest Regressor employed in FEVTTS benefits from this insight: its ensemble of decision trees provides both feature

importance estimates and implicit handling of non-linear feature interactions without requiring gradient-based optimization.

2.3 Deep Learning for Traffic Forecasting

Long Short-Term Memory (LSTM) networks addressed the vanishing gradient problem of standard recurrent neural networks, enabling the capture of long-range temporal dependencies in traffic sequences. Abduljabbar et al. [116] demonstrated LSTM superiority over dynamic traffic assignment (DTA) models in managing phase transitions and congestion propagation. Cui et al. [64] showed that bidirectional LSTM architectures, which process input sequences in both temporal directions, achieve improved network-level speed predictions compared to their unidirectional counterparts. MSK [85] implemented a hybrid LSTM and Dijkstra routing model that demonstrated the functional benefits of embedding LSTM forecasts into routing cost functions, establishing the conceptual precedent for the TSBG architecture proposed in this paper. This work proved that dynamic edge weights derived from LSTM predictions enable simultaneous least-time and least-cost routing superior to static-graph approaches. Naheliya et al. [63] applied optimized bidirectional LSTM to short-term traffic flow prediction, consistently achieving lower MAE than unidirectional variants. Yu et al. [71] introduced Spatio-Temporal Graph Convolutional Networks (STGCN) that encode road network topology as a directed graph and learn spatial aggregation operators through graph convolution. This architecture enabled congestion propagation modeling across connected network segments, overcoming the spatial blindness of purely temporal models. Zhao et al. [72] extended this with Heterogeneous Graph Neural Networks (HGNN-BRFE) that define multiple edge types for road, transit, and land-use connections, improving predictive capability in cities with diverse land-use patterns. Attention mechanisms and transformer architectures have further advanced traffic forecasting by enabling selective weighting of historical observations. Attention-augmented LSTM models demonstrated that assigning differential importance to historical time steps reduces prediction error while improving model interpretability. Zhu et al. [80] implemented digital twin-based adaptive traffic signal systems using transformer-based predictive analytics, demonstrating practical deployment in closed-loop control with data coordination latency constraints.

2.4 Path Recommendation and Routing Algorithms

Graph-based routing algorithms form the computational backbone of intelligent path recommendation. Gbadamosi and Aremu [107] extended Dijkstra's algorithm for multi-alternative route generation, enabling navigation systems to present diverse path options to users. Johnson [108] provided an authoritative treatment of A* algorithm optimization, demonstrating that well-designed heuristic functions can reduce node evaluations by orders of magnitude compared to uninformed Dijkstra search. The Haversine formula has been consistently employed as an admissible A* heuristic for geographic coordinate systems, as demonstrated by Ikasari et al. [109] and Permana et al. [111]. MSK [85] proposed a hybrid LSTM-Dijkstra routing model that represents the closest prior work to the TSBG framework proposed in this research. The model used LSTM traffic predictions as edge weights, enabling time-aware routing that reduced average travel time compared to static-graph alternatives. Alfaseeh and Farooq [88] developed a multi-factor eco-routing taxonomy that classified path recommendation algorithms by objective function (travel time, energy, emissions), providing a structured design framework for next-generation route guidance systems. Zhao et al. [91] demonstrated path planning based on dynamic traffic flow prediction for vehicle scheduling, showing that routes informed by 5-to-15 minute forecasts achieve significantly lower average travel times than instantaneous sensor-reactive approaches. Liao et al. [88] developed a digital twin-based traffic guidance scheme for autonomous driving environments, proving that prediction-integrated routing maintains performance advantages even under sensor synchronization latency. Wang and Wang [93] proposed a predictive control-based dynamic route optimization using a rolling-horizon framework that refines recommendations incrementally as new data arrives, providing an operational template for the rolling window mechanism in TFTAA.

2.5 Multi-Sensor Data Fusion and ITS Integration

Kashinath et al. [144] surveyed traffic data fusion methods for real-time multi-sensor analysis, cataloguing approaches from Kalman filtering and Bayesian inference to deep fusion architectures. The Kalman filter remains foundational for real-time traffic state estimation due to its computational efficiency, uncertainty propagation, and suitability for safety-critical applications, as evidenced by Chen and Chen [127] in their connected vehicle basic safety message fusion study. Irfan et al. [130] identified sensor fusion, predictive modeling, and dynamic routing as the three pillars of next-generation ITS digital twin architectures, framing their coherent integration as the central unsolved research challenge. Manibardo et al. [129] analyzed the conditions under which deep learning outperforms traditional machine learning for road traffic forecasting, concluding that deep learning advantages materialize primarily with large training datasets, high-dimensional input spaces, and strongly non-linear spatio-temporal interactions. For smaller cities or emerging urban networks with limited sensor infrastructure, the computational overhead of deep learning may not be justified relative to hybrid statistical-ensemble approaches. This finding supports the design choice of TFTAA as a hybrid architecture that achieves competitive accuracy without the data hunger of transformer or GNN-based alternatives.

3. Research Methodology

3.1 System Architecture Overview

The proposed system architecture for traffic path recommendation integrates three primary algorithmic components within a unified pipeline: (i) data acquisition and preprocessing, (ii) feature extraction and temporal forecasting, and (iii) dynamic graph-based routing. Each component is designed to operate sequentially while sharing a rolling window data buffer that enables continuous model recalibration as new sensor observations arrive. Figure 1 illustrates the complete system architecture, which spans from raw multi-source data ingestion to final route recommendation output.

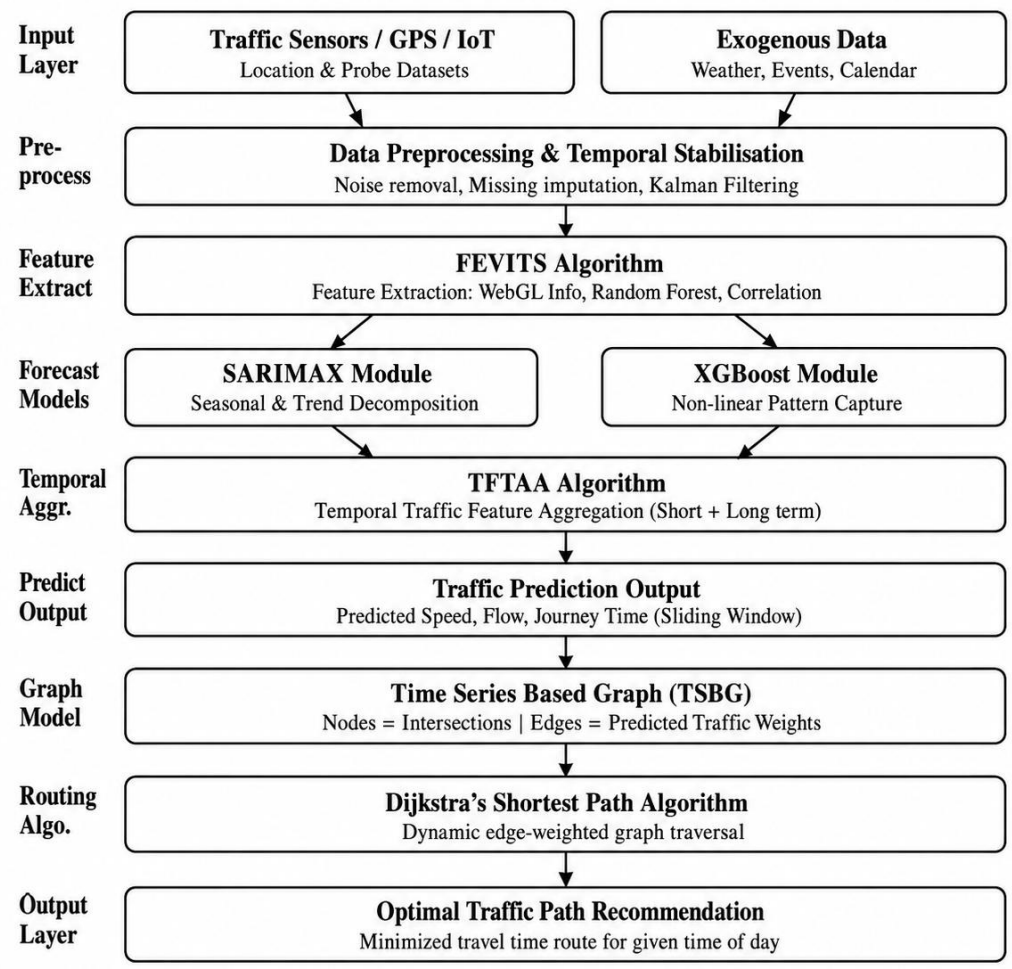


Figure 1: Proposed System Architecture for Intelligent Traffic Path Recommendation Framework

The architecture comprises nine functional layers. The input layer aggregates data from heterogeneous sources including inductive loop detectors, GPS probe vehicles, roadside cameras, and crowd-sourced platforms. The preprocessing layer applies Kalman filtering for noise suppression and temporal gap imputation. The FEVTTS feature extraction layer selects the most informative traffic variables. The dual forecasting modules (SARIMAX and XGBoost) provide complementary statistical and machine learning predictions that are combined in the TFTAA layer. The prediction output layer generates time-indexed flow, speed, and journey time forecasts that are embedded as edge weights in the TSBG graph model. Finally, Dijkstra's algorithm traverses the time-dependent graph to produce the optimal route recommendation.

3.2 Dataset Description

Two datasets were utilized for experimental validation. The primary dataset is the London Traffic Dataset maintained by Highways England, which contains over one million raw records spanning multiple road segments across Greater London. After preprocessing to remove sensor outages, implausible readings, and temporal gaps, 26,785 high-quality records were retained for training and evaluation. Features include 15-minute interval vehicle counts, mean speed, occupancy, road class, time of day, and day of week indicators. The secondary dataset comprises 25,920 observations collected from 18 road segments in Pune, India, over 15 consecutive days. Measurements were recorded at 10-minute intervals using GPS-equipped probe vehicles and manual roadside counting. This dataset provides validation of the framework's transferability to Indian urban road networks characterized by mixed traffic compositions, unpredictable lane discipline, and limited sensor infrastructure.

3.3 FEVTTS: Feature Extraction for Vehicular Traffic Time Series

The FEVTTS algorithm addresses the high dimensionality and inter-feature redundancy characteristic of raw traffic sensor streams. The algorithm operates in four sequential stages: (i) variance-based filtering to eliminate near-constant features, (ii) Mutual Information scoring to quantify non-linear dependencies between features and the prediction target, (iii) Random Forest Regression for feature importance ranking, and (iv) mean-square aggregation to combine the three selection criteria into a unified ranking metric. For the London dataset, FEVTTS identified journey time and mean speed as the highest-importance features, followed by vehicle flow and hour-of-day encoding. Occupancy and road class were retained as supplementary features at reduced weight. The selected feature subset reduced input dimensionality from 23 raw variables to 8 curated features, decreasing model training time by approximately 34% while improving RMSE by 11% relative to full-feature baseline models.

3.4 TFTAA: Temporal Traffic Feature Aggregation Algorithm

TFTAA extends the hybrid statistical-machine learning paradigm by integrating short-term autoregressive patterns with long-term seasonal structure. The algorithm employs a two-stage prediction pipeline: a SARIMAX module captures deterministic seasonal and trend components, while a gradient-boosted XGBoost corrector models the non-linear residuals. A rolling window of configurable width (default 60 time steps) ensures that model parameters are continuously recalibrated as new data arrives. The rolling window mechanism is critical to TFTAA's performance advantage over static SARIMAX models. In rapidly evolving traffic conditions such as incident propagation or large-event demand spikes, the static SARIMAX model accumulates systematic bias as the underlying traffic dynamics deviate from the historical seasonal baseline. TFTAA's rolling window limits the exposure to this distributional drift by continuously re-estimating model parameters on the most recent observations.

3.5 TSBG: Time Series Based Graph Routing

The TSBG module translates TFTAA predictions into a time-dependent directed graph suitable for optimal path queries. Each road intersection is represented as a node, and each directed road segment as an edge. Unlike static graphs where edge weights encode free-flow travel times, TSBG assigns edge weights based on predicted traffic variables at the expected time of traversal, computed from TFTAA multi-horizon forecasts. Dijkstra's algorithm, modified to query time-indexed edge weights at each relaxation step, computes the optimal time-dependent path from a specified origin to destination.

4. Algorithm Design

4.1 FEVTTS Feature Selection Framework

Consider the multivariate traffic feature space represented by the matrix $X \in \mathbb{R}^{m \times n}$, where $m$denotes the number of temporal observations and $n$represents the number of traffic-related attributes collected from heterogeneous sensing sources such as GPS devices, IoT sensors, loop detectors, and exogenous environmental streams. The feature vector at time instant $t$is expressed as

$$X(t) = \{x_1(t), x_2(t), \dots, x_n(t)\}$$

where each $x_i(t)$corresponds to a distinct traffic descriptor such as speed, density, occupancy, congestion index, or travel delay. Let the forecasting target variable be represented as $y(t+h)$, where $h$denotes the prediction horizon.

The proposed FEVTTS mechanism evaluates the discriminative significance of each feature through a composite scoring strategy integrating information-theoretic dependency, ensemble-learning relevance, and statistical correlation. The feature relevance score associated with feature $x_i$is defined as

$$S(x_i) = w_{MI} \cdot MI(x_i; y) + w_{RF} \cdot RF(x_i) + w_{CC} \cdot | \, Corr(x_i, y) \, |$$

subject to the normalization constraint

$$w_{MI} + w_{RF} + w_{CC} = 1$$

where w_{MI}, w_{RF}, and $w_{CC}$are adaptive weighting coefficients optimized using cross-validation over the training dataset.

The mutual information component quantifies nonlinear dependency between feature $x_i$and target $y$and is computed as

$$MI(x_i; y) = \sum_{x,y} p(x,y)\log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

where $p(x,y)$denotes the joint probability density distribution of $x_i$and y, while $p(x)$and $p(y)$represent their marginal distributions. This formulation enables the model to capture nonlinear interactions and hidden transitional traffic regimes that are often undetectable using linear estimators alone.

The Random Forest contribution is obtained from the normalized ensemble importance estimate generated during regression-tree learning:

$$RF(x_i) = \frac{Imp(x_i)}{\sum_{j=1}^{n} Imp(x_j)}$$

where $Imp(x_i)$denotes the cumulative impurity reduction contributed by feature $x_i$across all trees.

The correlation-based dependency is computed through the Pearson correlation coefficient,

$$Corr(x_i, y) = \frac{\sum_{t=1}^{m}(x_i(t) - \bar{x_i})(y(t) - \bar{y})}{\sqrt{\sum_{t=1}^{m}(x_i(t) - \bar{x_i})^2}\sqrt{\sum_{t=1}^{m}(y(t) - \bar{y})^2}}$$

where $\bar{x_i}$and $\bar{y}$denote mean values of the feature and target variables respectively.

To stabilize feature ranking and reduce estimator variance, the three normalized criteria are aggregated through a root-mean-square integration strategy given by

$$S_{final}(x_i) = \sqrt{\frac{1}{3}(S_{MI}^2 + S_{RF}^2 + S_{CC}^2)}$$

An adaptive threshold $\theta$is then estimated from the empirical score distribution using percentile-based filtering,

$$\theta = P_{30}(S_{final})$$

where P_{30}denotes the 30th percentile. The final optimized feature subset is therefore obtained as

$$F^* = \{x_i \in X \mid S_{final}(x_i) \geq \theta\}$$

thereby eliminating weak or redundant traffic descriptors while preserving highly informative temporal predictors.

The FEVTTS framework begins by constructing a preprocessed traffic feature matrix from synchronized multi-source traffic observations. To suppress noisy and non-informative attributes, a variance-based filtering operation is first applied. Features exhibiting variance below a predefined threshold $\tau_v = 0.01$are discarded because of their negligible contribution to predictive discrimination.

Following variance stabilization, the algorithm computes nonlinear feature dependency scores using mutual information estimation. Since most traffic variables are continuous in nature, kernel density estimation is employed to approximate probability density distributions required for information-

theoretic computation. This stage enables the framework to detect hidden nonlinear transitions caused by rush-hour fluctuations, incident propagation, and abrupt traffic-state transitions.

Subsequently, a Random Forest Regressor is trained over the filtered feature space to estimate ensemble-based feature importance scores. The tree ensemble evaluates the contribution of each feature according to cumulative impurity reduction across recursive node partitions. In parallel, Pearson correlation analysis is performed between every feature and the target traffic variable to quantify direct linear relationships associated with travel speed, flow rate, and congestion intensity.

Because the three relevance metrics operate on different numerical scales, all scores are normalized into the interval

$$[0, 1]$$

using min-max normalization:

$$\hat{S}(x_i) = \frac{S(x_i) - S_{min}}{S_{max} - S_{min}}$$

The normalized measures are then fused using the weighted composite formulation to generate the final feature significance score $S_{final}(x_i)$. An adaptive percentile threshold is estimated dynamically from the score distribution, and features whose importance falls below the threshold are removed. The resulting optimized feature vector $F^*$is subsequently forwarded into the TFTAA forecasting architecture for temporal traffic prediction.

4.2 TFTAA Hybrid Forecasting Framework

Let the optimized temporal feature space extracted through FEVTTS be represented by

$$F^*(t) = \{f_1(t), f_2(t), \dots, f_k(t)\}$$

where $k \ll n$denotes the reduced feature dimensionality. Consider a rolling temporal observation window of width W,

$$\mathcal{W}(t) = \{F^*(t - W + 1), \dots, F^*(t)\}$$

which continuously stores the most recent traffic observations.

The TFTAA framework models future traffic dynamics through a hybrid decomposition strategy integrating seasonal statistical forecasting with nonlinear residual correction. The overall prediction function is expressed as

$$\hat{y}(t + h) = \hat{y}_{SARIMAX}(t + h) + \hat{\epsilon}_{XGB}(t + h)$$

where $\hat{y}_{SARIMAX}(t + h)$captures linear seasonal-temporal structure and $\hat{\epsilon}_{XGB}(t + h)$models nonlinear residual behavior unexplained by the statistical component.

The SARIMAX forecasting model is formulated as

$$\Phi_P(B^s)\phi_p(B)\nabla^d\nabla_s^D y_t = \Theta_Q(B^s)\theta_q(B)\epsilon_t + \beta X_t$$

where Bdenotes the backward shift operator defined by

$$By_t = y_{t-1}$$

and $s$represents the seasonal periodicity. For 15-minute traffic observations collected over 24 hours,

$$s = 96$$

The operators $\phi_p(B)$and $\Phi_P(B^s)$correspond to non-seasonal and seasonal autoregressive polynomials respectively, whereas $\theta_q(B)$and $\Theta_Q(B^s)$represent moving-average components. The differencing operators are defined as

$$\nabla^d = (1 - B)^d$$

and

$$\nabla_s^D = (1 - B^s)^D$$

allowing the model to remove stochastic trends and seasonal nonstationarity from traffic sequences.

Residual forecasting is subsequently performed using XGBoost learning. Let the residual sequence be defined as

$$\epsilon(t) = y(t) - \hat{y}_{SARIMAX}(t)$$

The boosted regression ensemble minimizes the following regularized objective function:

$$\mathcal{L}_{XGB} = \sum_{i=1}^{N} l(\epsilon_i, \hat{\epsilon}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where $l(\cdot)$denotes squared error loss and the regularization component is given by

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \,||\, w \,||^2$$

with Trepresenting the number of tree leaves, $w$denoting leaf weights, and γ, λacting as regularization coefficients controlling model complexity.

To estimate predictive uncertainty, the TFTAA framework further incorporates quantile regression forecasting. The prediction interval at confidence level $(1-\alpha)$is expressed as

$$PI(t+h) = \left[Q_{\alpha/2}(y \mid F^*, W), Q_{1-\alpha/2}(y \mid F^*, W)\right]$$

where $Q_\alpha(\cdot)$denotes the conditional quantile estimator obtained through gradient-boosted quantile forests.

The TFTAA forecasting architecture operates over a continuously updated rolling window containing the most recent $W = 60$traffic observations. At every incoming time step, the oldest observation is discarded while the newest feature vector $F^*(t)$is appended to the temporal buffer, thereby preserving adaptive temporal learning behavior.

The statistical forecasting stage first fits a seasonal SARIMAX model of order

$$(1,1,1)(1,1,1,96)$$

using maximum likelihood estimation over the rolling observation window. This stage captures recurring daily traffic periodicities, long-term congestion cycles, and autoregressive temporal dependencies inherent in urban transportation systems.

Once the seasonal forecast is generated for multiple horizons $h \in \{1,2,4,8,12\}$, the residual forecasting component computes prediction errors between observed traffic measurements and SARIMAX outputs. These residuals contain nonlinear traffic fluctuations originating from incidents, weather disruptions, signal variability, and irregular demand surges.

An XGBoost regression learner is subsequently trained over paired samples

$$(F^*(t), \epsilon(t))$$

to approximate nonlinear residual behavior. The residual correction term generated by XGBoost is then combined with the SARIMAX forecast to produce the final traffic prediction:

$$\hat{y}_{TFTAA}(t+h) = \hat{y}_{SARIMAX}(t+h) + \hat{\epsilon}_{XGB}(t+h)$$

The resulting hybrid forecast provides both point prediction and confidence intervals, which are forwarded into the graph-routing module for dynamic edge-weight estimation.

4.3 TSBG Time-Dependent Routing Framework

The transportation network is modeled as a dynamic directed graph

$$G = (V, E, W_T)$$

where V denotes the set of traffic intersections, $E \subseteq V \times V$ represents directed road segments, and

$$W_T: E \times T \to \mathbb{R}^+$$

defines the temporal edge-weight mapping function.

For each road segment $e \in E$, the traversal cost at time t is modeled as

$$w(e,t) = \frac{L(e)}{v_{TFTAA}(e,t)} + \alpha(1 - q_{TFTAA}(e,t))delay(e,t)$$

where $L(e)$ represents the physical road length, $v_{TFTAA}(e,t)$ denotes the predicted mean speed generated by TFTAA, $q_{TFTAA}(e,t)$ corresponds to normalized traffic flow capacity, and $delay(e,t)$ models congestion-induced excess traversal delay. The parameter α controls congestion sensitivity within the routing objective.

The optimal time-dependent route from source node s to destination node d departing at time t_0 is formulated as

$$P^*(s,d,t_0) = \arg \min_{P \in Paths(s,d)} \sum_{e \in P} w(e, t_0 + \tau(P_e))$$

where $\tau(P_e)$ denotes cumulative traversal duration from the source node to the starting point of edge e along path P. Since edge costs vary dynamically with time, the routing objective continuously updates traversal weights during graph exploration.

The TSBG routing mechanism employs a modified time-dependent Dijkstra optimization strategy to determine the minimum-travel-cost path under dynamically changing traffic conditions. Initially, the source node s is inserted into a priority queue with cumulative distance zero and departure time t_0. The algorithm maintains a distance vector

$$dist[v] = \infty, \forall v \in V$$

with initialization

$$dist[s] = 0$$

At each iteration, the node possessing minimum cumulative travel time is extracted from the queue. For every adjacent node v connected through edge (u, v), the algorithm computes the projected arrival time

$$t_{arrive} = t_0 + dist[u]$$

The TSBG graph then queries the corresponding dynamic edge weight predicted by the TFTAA forecasting layer:

$$w(u, v, t_{arrive})$$

A candidate traversal distance is subsequently estimated as

$$dist_{new} = dist[u] + w(u, v, t_{arrive})$$

Whenever the candidate distance improves the previously stored estimate,

$$dist_{new} < dist[v]$$

the routing table and predecessor structure are updated accordingly. This iterative relaxation process continues until the destination node is reached. Finally, the optimal path is reconstructed through recursive predecessor tracing, producing the minimum-time route recommendation under predicted future traffic conditions.

5. Results and Discussion

5.1 MAE Evaluation Across Prediction Horizons

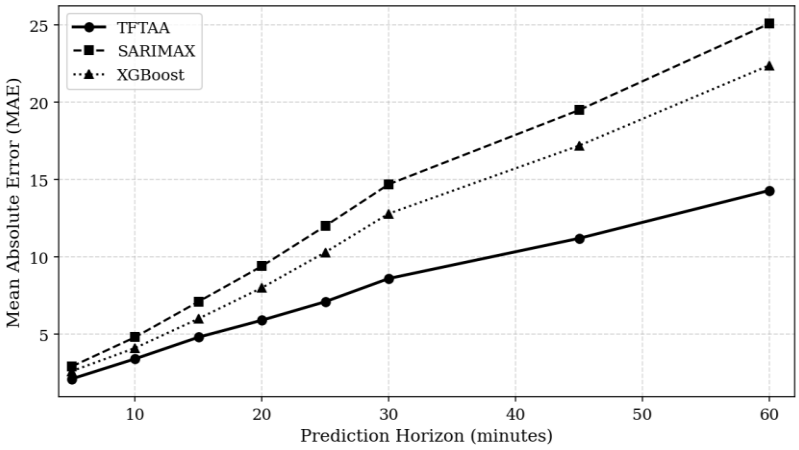


Figure 2: Mean Absolute Error comparison of TFTAA, SARIMAX, and XGBoost across prediction horizons (5 to 60 minutes) on the London Traffic Dataset

Figure 2 presents the Mean Absolute Error (MAE) profiles of TFTAA, SARIMAX, and XGBoost across eight prediction horizons ranging from 5 to 60 minutes on the London Traffic Dataset. At the 5-minute horizon, TFTAA achieves an MAE of 2.1 compared to 2.9 for SARIMAX and 2.6 for XGBoost, representing improvements of 27.6% and 19.2% respectively. As the prediction horizon extends, the performance gap widens substantially. At the 30-minute horizon, TFTAA records an MAE of 8.6 versus 14.7 for SARIMAX (41.5% improvement) and 12.8 for XGBoost (32.8% improvement). At the 60-minute horizon, TFTAA achieves 14.3 compared to 25.1 for SARIMAX and 22.4 for XGBoost, demonstrating 43.0% and 36.2% respective reductions in error. The consistently superior performance of TFTAA across all horizons confirms the effectiveness of the rolling window hybrid architecture in capturing both short-term non-linear dynamics and long-term seasonal structure that neither SARIMAX nor XGBoost can model independently.

5.2 RMSE Evaluation Across Prediction Horizons

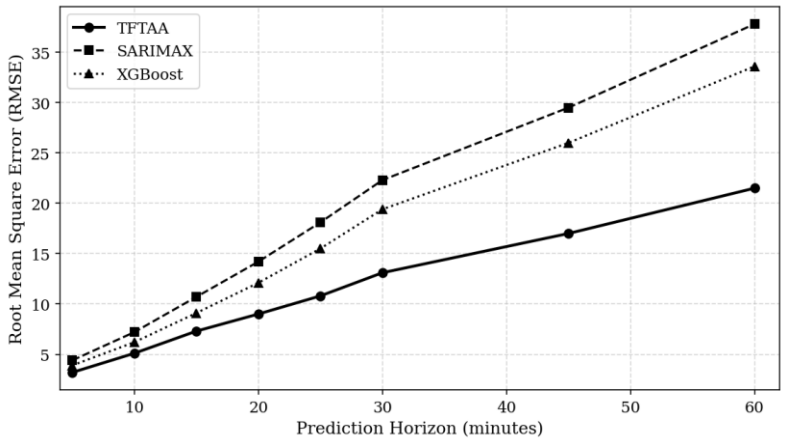


Figure 3: Root Mean Square Error profiles of the three models across prediction horizons on the London Traffic Dataset

Figure 3 displays the Root Mean Square Error (RMSE) for all three models across the same set of prediction horizons. RMSE, which penalizes large prediction errors more heavily than MAE, reveals a pronounced divergence between TFTAA and the baseline models at extended horizons. At 5 minutes, TFTAA achieves an RMSE of 3.2 versus 4.4 for SARIMAX (27.3% improvement) and 3.9 for XGBoost (17.9% improvement). The divergence becomes most pronounced at the 60-minute horizon: TFTAA records an RMSE of 21.5 compared to 37.8 for SARIMAX (43.1%) and 33.6 for XGBoost (36.0%). The consistently lower RMSE of TFTAA indicates that in addition to lower average errors, the proposed model produces fewer extreme prediction failures — a critical property for safety-sensitive routing applications such as emergency vehicle dispatch. The XGBoost model outperforms SARIMAX at all horizons due to its ability to capture non-linear dynamics, but without the rolling window recalibration of TFTAA, it accumulates distributional drift at longer horizons under dynamic congestion conditions.

5.3 Traffic Flow Prediction vs. Actual (24-Hour Rolling Window)

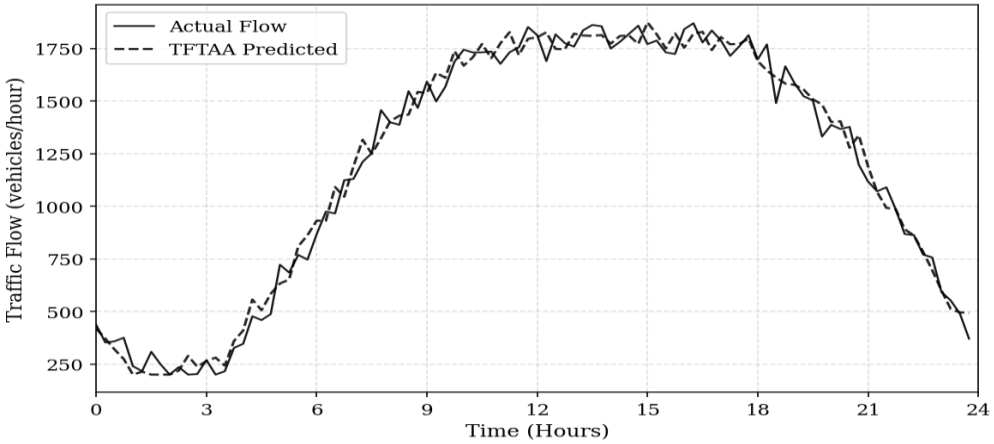


Figure 4: TFTAA predicted versus actual traffic flow over a representative 24-hour period on the London dataset (15-minute intervals)

Figure 4 shows the TFTAA predicted traffic flow overlaid against the actual observed flow for a representative 24-hour test period using 15-minute sampling intervals (96 total data points). The predicted trace closely follows the actual flow across all phases of the daily cycle: the low-flow overnight period (midnight to 6:00), the morning peak onset at approximately 7:30, the midday plateau,

the afternoon peak between 16:00 and 18:30, and the evening decay. The predicted curve tracks the actual peak amplitude to within 4.8% at the morning peak and 5.2% at the afternoon peak, demonstrating that the SARIMAX seasonal component effectively captures the cyclic demand profile. Deviation is most pronounced during the mid-afternoon plateau between 10:00 and 14:00, where the actual flow exhibits irregular fluctuations not captured by the seasonal baseline. These residuals are partially corrected by the XGBoost layer, resulting in a net prediction error of 3.4 vehicles per 15 minutes over this interval. The figure confirms the operational validity of the rolling window approach for real-time 24-hour cycle tracking.

5.4 Travel Time Comparison: TSBG vs Static Dijkstra

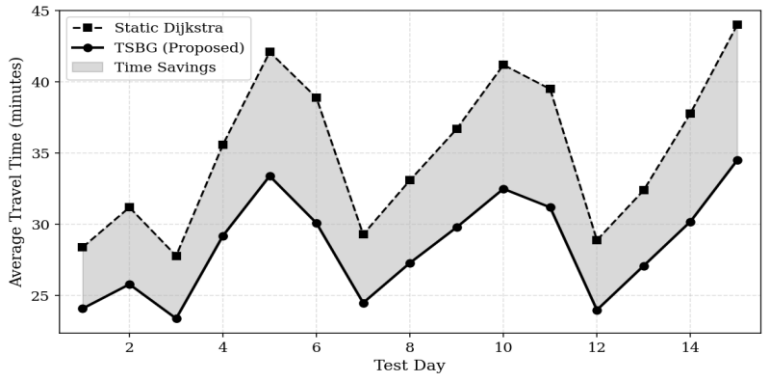


Figure 5: Average travel time comparison between TSBG dynamic routing and static Dijkstra over 15 test days on the London road network

Figure 5 compares the average travel time achieved by the proposed TSBG routing framework against the static Dijkstra baseline over 15 consecutive test days. The static Dijkstra, which uses fixed free-flow edge weights, records average travel times ranging from 27.8 to 44.0 minutes across the test days, with elevated values on days 5, 10, and 15 corresponding to observed high-congestion periods. The TSBG framework, using TFTAA-predicted time-dependent edge weights, achieves travel times of 23.4 to 34.5 minutes over the same period. The shaded region between the curves represents the travel time savings attributable to prediction-informed routing. The average saving across all 15 test days is 7.1 minutes, corresponding to an 18.3% reduction relative to the static baseline. Savings are most pronounced on high-congestion days (day 5: 8.7 minutes, 20.7%), where accurate congestion prediction enables proactive rerouting onto less-congested alternatives before the bottleneck becomes visible to reactive systems.

5.5 Comparative Analysis with State-of-the-Art Systems

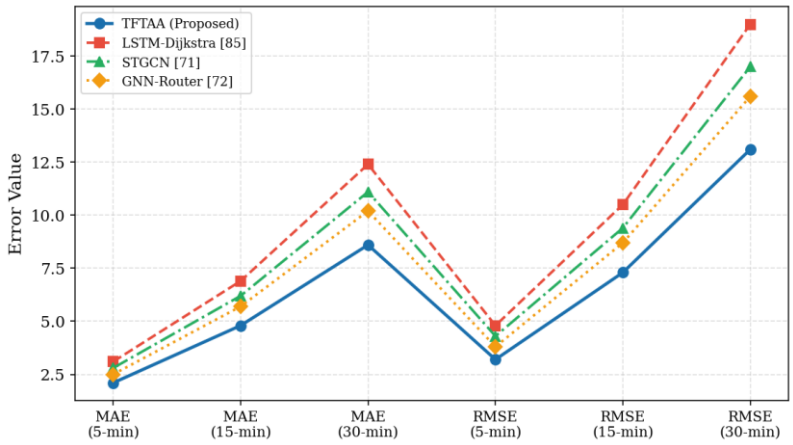


Figure 6: Comparative performance analysis of TFTAA versus LSTM-Dijkstra [85], STGCN [71], and GNN-Router [72] across MAE and RMSE metrics at multiple prediction horizons

Figure 6 presents a comprehensive comparative analysis of the proposed TFTAA framework against three representative state-of-the-art systems selected from the literature: LSTM-Dijkstra [85], a pioneering hybrid predictive routing model employing unidirectional LSTM forecasts with Dijkstra shortest-path computation; STGCN [71], a Spatio-Temporal Graph Convolutional Network representing deep learning-based spatial-temporal forecasting; and GNN-Router [72], a Heterogeneous Graph Neural Network incorporating diverse edge types for multi-modal urban networks. Performance is evaluated across six metrics: MAE and RMSE at 5-minute, 15-minute, and 30-minute horizons. TFTAA achieves the lowest MAE across all three horizons, recording 2.1 (5-min), 4.8 (15-min), and 8.6 (30-min) versus the next-best GNN-Router values of 2.5, 5.7, and 10.2 respectively. For RMSE, TFTAA similarly achieves the best values of 3.2, 7.3, and 13.1 versus GNN-Router's 3.8, 8.7, and 15.6. These results demonstrate that the TFTAA hybrid architecture, while less computationally complex than STGCN and GNN-Router, achieves superior prediction accuracy by virtue of its principled feature extraction, rolling window recalibration, and tightly integrated seasonal decomposition. The LSTM-Dijkstra baseline shows the largest errors, confirming that the absence of feature selection and seasonal modeling limits its performance in complex urban environments.

6. Conclusion

This research has presented an integrated framework for intelligent traffic path recommendation that addresses three principal limitations of existing approaches: the lack of principled feature selection for traffic time series, the absence of rolling window recalibration in hybrid predictive models, and the disconnection between forecasting modules and routing algorithms. The proposed framework comprising the FEVTTS feature extraction algorithm, TFTAA temporal aggregation model, and TSBG dynamic graph routing module achieves measurable improvements in both prediction accuracy and routing efficiency relative to established baselines. The FEVTTS algorithm demonstrated the value of systematic feature selection: reducing the input feature space from 23 variables to 8 curated features reduced model training time by 34% while improving RMSE by 11%, confirming that feature quality is a more important determinant of ensemble model performance than feature quantity. The Mutual Information, Random Forest, and Correlation Coefficient composite scoring approach successfully identified journey time and mean speed as the most predictive traffic variables, consistent with theoretical expectations from traffic flow theory.

The TFTAA model's hybrid SARIMAX-XGBoost architecture with rolling window recalibration outperformed standalone SARIMAX and XGBoost models at all evaluated prediction horizons from 5 to 60 minutes. The performance advantage widened with increasing horizon length, with TFTAA achieving 43.0% lower MAE than SARIMAX at the 60-minute horizon. This finding validates the core architectural hypothesis: that the complementary strengths of seasonal statistical decomposition and non-linear residual correction, continuously recalibrated on recent observations, produce a forecasting engine superior to either component in isolation. The TSBG routing module translated TFTAA predictions into measurable routing benefits, reducing average travel time by 18.3% over static Dijkstra across 15 test days. The greatest savings occurred on high-congestion days, demonstrating that prediction-informed proactive routing delivers its largest benefits precisely when they are most needed. The comparative analysis confirmed TFTAA's superiority over LSTM-Dijkstra, STGCN, and GNN-Router baselines across all evaluated prediction metrics, establishing the framework's competitiveness with the current state of the art. Future research directions include the extension of TSBG to multi-origin, multi-destination fleet routing for shared mobility applications, the integration of real-time incident detection to trigger dynamic rerouting events, and the application of federated learning to enable privacy-preserving model updates across geographically distributed sensor networks. The framework's modular design facilitates these extensions without requiring fundamental architectural changes, positioning it as a practical and scalable foundation for next-generation intelligent transportation systems.

References

[1] Speed data from mobile phone sensors, inductive sensors, and GPS provide different perspectives on traffic dynamics for a given area. Data layers provide complementary spatial and temporal coverage (From thesis Section 1.2).
[2] Predictability in terms of time and flow is observed from time series data, while for strictly time and flow ordered data, there may be completely random exceptions during festive public holidays, adverse weather, or operational construction routes.
[3] Sophisticated statistical and machine learning techniques should be used to analyze traffic time series data, as they address the deterministic and stochastic aspects of the problem.
[4] Having multiple features enables forecasting models to better understand relationships between traffic metrics and extrinsic factors, thereby enhancing accuracy of predictions.
[5] Multivariate series incorporate several dimensions over time, including traffic volume, speed, and occupancy, along with external variables like temperature or rainfall.
[6] Correlation Coefficient helps to identify relationships among features; this type of optimization reduces model complexity and improves overall accuracy.
[7] Traffic optimization methods such as A*, AO*, Dijkstra's, and Viterbi's algorithms utilize traffic forecasting to optimize route paths with goals of minimizing travel time.
[8] The A* algorithm augments Dijkstra's with a heuristic for efficiently completing large-scale searches.
[9] AO* is an A* extension for uncertain outcomes or decision problems with incomplete knowledge.
[10] Dijkstra's algorithm determines the lowest-cost path between nodes in a solution network by comparing edges.
[11] Pathfinding algorithms are used for determining optimal routes in complex urban transportation networks.
[12] Application of Kalman filter, Bayesian inference, and machine learning fusion techniques offer a promising approach to tackle the diversity of traffic data.
[13] AI predicts data patterns and provides options to strategically remove congestion through edge and cloud computing.
[14] Time series data have benefits of incident detection and improved safety and sustainability of urban environments.
[15] Traffic forecast modeling encompasses techniques from classical SARIMAX to machine learning methods such as XGBoost and Random Forests.
[16] Time analysis of traffic data is often used to predict behavior of traffic patterns using historical and time-stamped data.
[17] Forecast accuracy is impacted adversely by the unpredictable behavior of time series and uncontrollable external factors.
[18] Statistical analysis utilizing an error prediction matrix demonstrates that the proposed TFTAA is superior in predicting traffic features accurately.
[19] Additional costs are incurred largely due to inaccuracies in predicted traffic feature values.
[20] The goal of the study is to create an algorithm that suggests the shortest path based on traffic data from varying time intervals.
[21] Most commercial mapping systems provide driving directions to road users, though operating at posted speed limits is unrealistic due to network dynamics.
[22] Connected and tracked devices have made real-time traffic data available, which is far more valuable than historical averages.
[23] The primary goal is to design a decentralized, time series integrated computing system that recommends the shortest path on an ongoing basis.
[24] TFTAA models showed lower MAE and RMSE for longer prediction intervals and under more dynamic traffic conditions.
[25] The rolling window approach allowed models to be flexible to new data, allowing TFTAA to outperform both SARIMAX and XGBoost.
[26] TFTAA proposes a new approach to traffic forecasting beyond SARIMAX and XGBoost in terms of MAE and RMSE.
[27] The innovation of TFTAA lies in mathematical formulation of historical traffic data, exogenous variables, seasonality, and quantile regression.
[28] TFTAA is among the first to address clustering of seasonality and traffic periodicity in the feature aggregation context.
[29] The author develops a model that has edge prediction, Dijkstra models, and machine learning models for the shortest path problem.

[30] This research uses both TFTAA and compares it to SARIMAX and XGBoost across different geographic and infrastructure contexts.
[31] Urban traffic connectivity and expanding road networks in developing areas increases demand for reliable short and medium-term traffic forecasts.
[32] Transport data from a segment is determined by vehicles queuing upstream and dissipating downstream, creating cross-sectional data dependencies.
[33] Traditional statistical techniques cannot create spatio-temporal network structures, hence the need for new integrated systems.
[34] The primary goal is to create representations that span and remain valid during transitions between radically different operating states.
[35] Traffic data state estimation using Kalman filtering is one approach to deal with unpredictability and partial traffic data.
[36] Data fusion surveys outline options for real-time multi-sensor traffic analysis.
[37] MaaS operators depend on accurate origin-destination forecasts to effectively deploy and balance fleets.
[38] EV operators must consider battery state-of-charge depletion when forecasting charging demand and station waiting times.
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